Database Systems Implementation
Mini-Project

1075201

Computer Science (Part C)

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In all pseudocode, we omit logging and detailed error handling for clarity; these details are implemented in the codebase.

1

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i

We manage tuple-level locks using shared/exclusive locks (`XSLocks` in Figure 1). These are standard two-phase locks: they operate in shared mode for queries and exclusive mode for edits (add/delete).

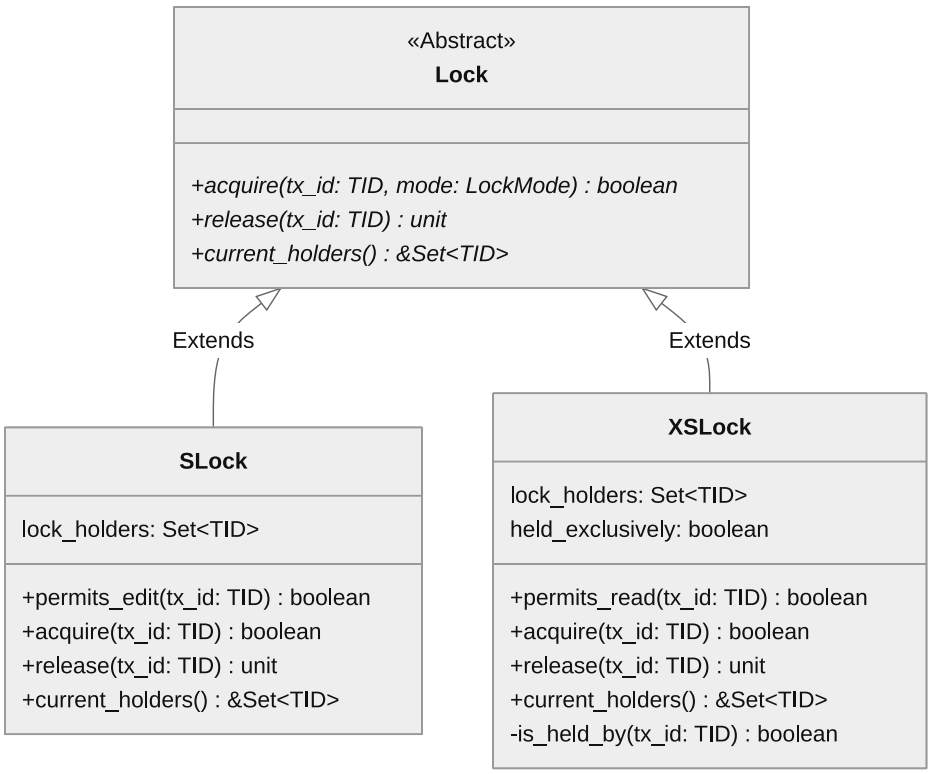

Figure 1: Lock Hierarchy

XSLocks are directly incorporated into the `DataTuple` class (Figure 2). The lock maintains an explicit set of `lock_holders` (`std::flat_set`) to support the deadlock detector's traversal of the waits-for graph. We use a flat set because the number of concurrent holders for a single tuple is typically very small, and a flat set improves cache locality for small sets.

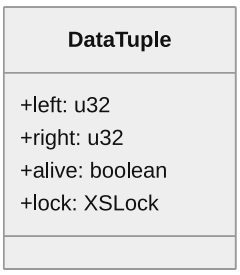

Figure 2: DataTuple Class

We use inheritance and a vtable to implement the lock hierarchy, rather than a C++20 concept. This allows each transaction to hold sets of `Lock` pointers and treat them uniformly.

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To support serializable isolation and prevent phantoms, we implement predicate locks using shared locks (`SLocks` in Figure 1). These locks represent conditions of the form "all tuples matching predicate P".

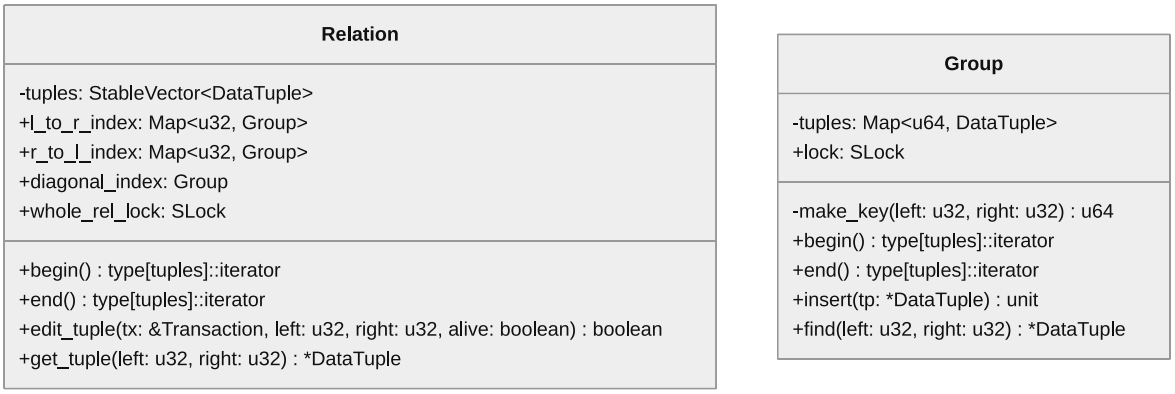

Figure 3: Relation and Group Classes

We provide two levels of granularity for predicate locks:

1. Relation Lock: Covers all tuples in a relation $R(x, y)$.
2. Group Locks: Cover subsets of a relation based on indices. These handle conditions of the form $R(c, x)$, $R(x, c)$, and $R(x, x)$ (diagonal).

Relation locks belong directly to `Relation` objects, while group locks are stored in `Groups` belonging to relations (see Figure 3). The group locks are organised into three separate groups: one for the left attribute, one for the right attribute, and one for the diagonal (where left == right). We will later see in 1(c) how these granularities of locks neatly cover the types of queries we have to evaluate.

The different operations interact with the locks as follows:

Queries read ranges of tuples matching a predicate. For example, a query $R(1, Y)$ requires reading all tuples where the left attribute is 1 in relation R. To prevent phantoms, we acquire a shared lock on the relevant group lock (e.g., the group lock for left attribute 1). This ensures that if another transaction tries to insert a tuple that matches this predicate, it is blocked until the query transaction commits or rolls back.

While reading existing tuples, we also need to prevent dirty reads from transactions that have not yet committed (or have aborted). Thus, queries call `Transaction::get_read_permit`, which checks `XSLock::permits_read` and records the tuple lock in `required_locks` when permission is denied.

There is an edge case for queries of the form $R(c_1, c_2)$, i.e. constant queries. For these queries, we do not take any predicate lock covering the (c_1, c_2) tuple in R since there is no risk of phantoms. On the other hand, we still need to prevent modifications to the tuple and thus must acquire the tuple-level lock in shared mode rather than only using `XSLock::permits_read`.

Edit operations modify tuples from an explicit list. As they do not use implicit ranges, they do not need to acquire predicate locks directly[1]. However, they still need to avoid causing phantoms, so for every tuple they edit, they must ensure that no other transaction has a shared lock on the relevant group lock covering that tuple's predicate. This is achieved using the `SLock::permits_edit` method, which checks whether the lock could be acquired in exclusive mode by this transaction. When editing a tuple, a transaction also acquires an exclusive lock on the tuple itself, as in standard two-phase locking.

[1]This is why we use `SLocks` rather than `XSLocks` for predicate locks.

All these interactions are summarised in Table 1.

	Predicate Locks	**Tuple Locks**
Type	SLock	XSLock
Granularity	Relation/Group	Single tuple
Interaction with Edits	Checked for edit permissions	Acquired in exclusive mode
Interaction with non-constant queries	Acquired in shared mode	Checked for read permissions
Interaction with constant queries	Not applicable	Acquired in shared mode

Table 1: Summary of Locks and their Interactions with Operations

Apart from the lock objects and their owners (`Group`, `Relation` and `DataTuple`), we also describe the rest of the lock management system, starting with `Transaction` objects (Figure 4) which hold onto pointers to locks.

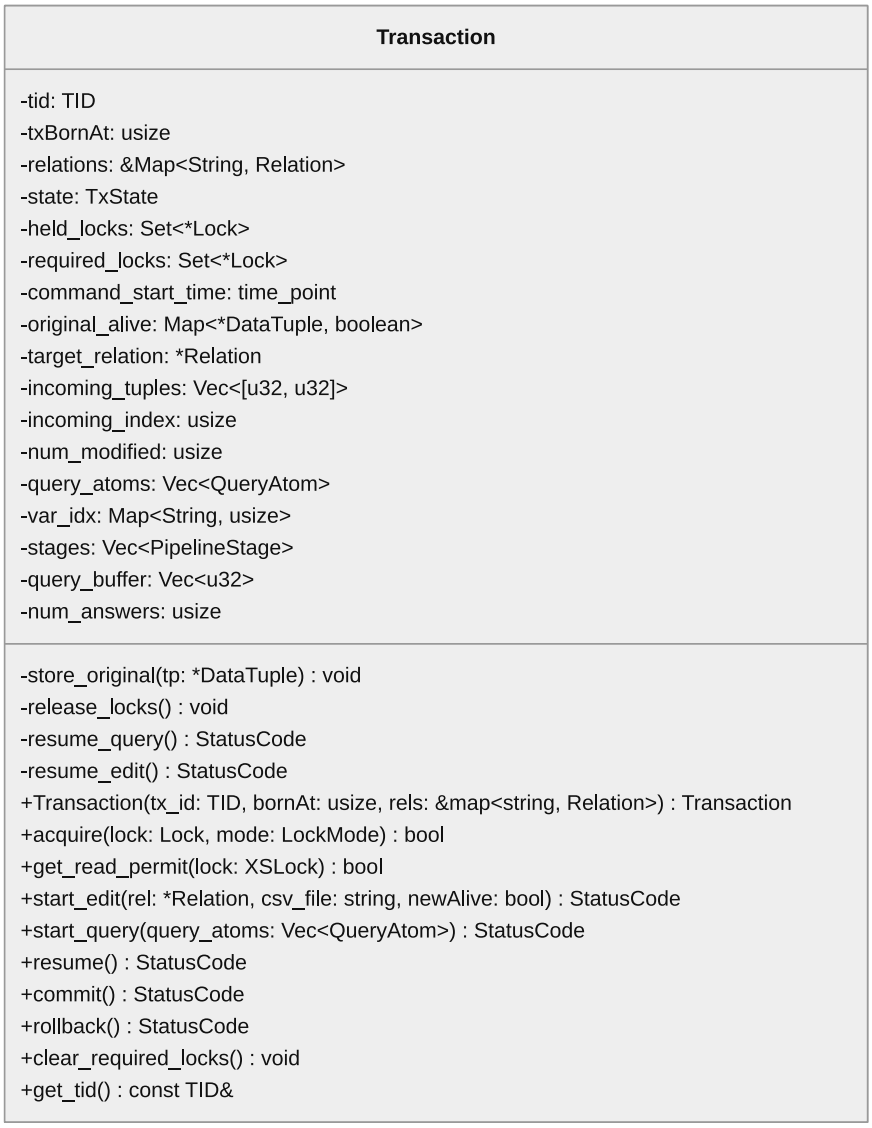

Figure 4: Transaction Class

We track both the locks held by each transaction (to release them on commit/rollback) and the locks a transaction is waiting on (for deadlock detection). A transaction likely holds many locks, so we use a hash set[2] to store the locks held by each transaction. On the other hand, we expect the number of locks that a transaction is waiting on to be small, so we use a `std::flat_set` to store the locks that a transaction is waiting on.

These transactions are stored in a single `Database` object (Figure 5) that acts as the global transaction and lock manager. When transactions finish an operation, they are either suspended or ready to continue. The `Database` class is responsible for detecting deadlocks in the former case and progressing normal control flow in the latter case. This is implemented in `Database::on_control`.

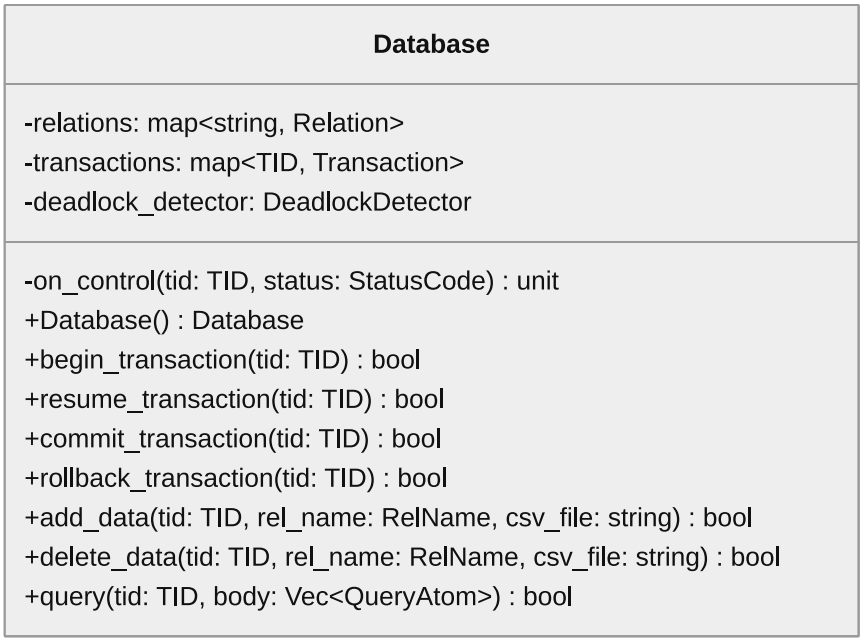

Figure 5: Database Class

We move some of the deadlock detection logic into a separate `DeadlockDetector` class (Figure 6), composed into the `Database` class. This keeps the core transaction/locking management logic separate from deadlock detection.

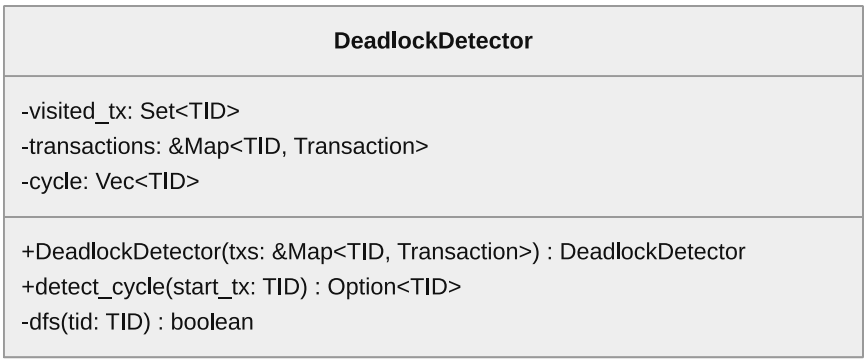

Figure 6: Deadlock Detector Class

iii

The locks expose the following primary functions:

[2]We write a custom open-addressing hash set for performance, see later discussion

`Lock::acquire(tx_id: TID, mode: LockMode) -> boolean`
Attempts to register the transaction as a holder. For XSLocks, this includes checking for shared/exclusive compatibility (see Algorithm 1). If a conflict occurs (e.g., requesting EXCLUSIVE when another transaction holds the lock), it returns `false`. For SLocks, we can always add the transaction as a holder and return `true`.

`Lock::release(tx_id: TID)`
Removes the transaction from the holder set. This is called during commit or rollback via `Transaction::release_all_locks()`, which both releases locks and clears the transaction's `held_locks` set.

`Lock::current_holders() -> &Set<TID>`
Returns the transactions currently holding the lock, used by the deadlock detector to construct the waits-for graph.

`SLock::permits_edit(tx_id: TID) -> boolean`
Checks whether the transaction can acquire the lock in exclusive mode, used by edit operations to prevent phantoms. This returns `false` if there is another transaction holding the lock (apart from `tx_id` itself), and `true` otherwise.

`XSLock::permits_read(tx_id: TID) -> boolean`
Checks whether the transaction can acquire the lock in shared mode, used by query operations to prevent dirty reads. This returns `false` if there is another transaction holding the lock in exclusive mode (apart from `tx_id` itself), and `true` otherwise.

Algorithm 1: XSLock::Acquire

```
procedure XSLock::Acquire(tx_id: TID, mode: LockMode) → boolean
  if mode = EXCLUSIVE then
    if holder_count() >= 2 then
      return false
    else if holder_count() = 1 and ¬ is_held_by(tx_id) then
      return false
    else
      # Upgrade or Acquire
      lock_holders.insert(tx_id)
      held_exclusively ← true
      return true
    end if
  else
    if held_exclusively then
      return is_held_by(tx_id)
    else
      lock_holders.insert(tx_id)
      return true
    end if
  end if
end procedure
```

The transaction objects expose the following primary functions related to lock management:

`Transaction::acquire(lock: Lock, mode: LockMode) -> boolean`
A wrapper around `Lock::acquire` used by the execution engine. If `Lock::acquire` fails, the lock is inserted into the transaction's `required_locks` set. The function returns `false` to signal that the transaction must be suspended.

`Transaction::get_read_permit(lock: XSLock) -> boolean`
Checks whether the transaction is allowed to read a tuple lock (`lock.permits_read(tid)`). If not, the lock pointer is inserted into `required_locks` before returning `false`, so suspended query stages expose the correct waits-for dependencies.

`Transaction::release_all_locks() -> unit`
Releases all locks held by the transaction and clears the `held_locks` set. This is called when a transaction commits or rolls back.

The deadlock detector object provides the following function:

`DeadlockDetector::detect_cycle(tid: TID) -> Optional<TID>`
Performs a depth-first search starting from the given transaction ID, traversing the waits-for graph by following locks that the transaction is waiting on and the transactions currently holding those locks. If we encounter a back edge, we have found a cycle. We then locate the youngest transaction in the cycle and return it as the victim to roll back. This is described in more detail in Algorithm 2.

Algorithm 2: Deadlock Detection

```
procedure DeadlockDetector::detect_cycle(start_tid: TID) → Optional<TID>
    visited_tx.clear()
    cycle.clear()
    if DFS(start_tid) then
        # Locate the youngest transaction in the cycle to abort
        victim ← cycle[0]
        for each tid in cycle do
            if transactions[tid].born_at > transactions[victim].born_at then
                victim ← tid
            end if
        end for
        return victim
    end if
    return empty
end procedure
procedure DeadlockDetector::DFS(tid: TID) → boolean
    if visited_tx[tid] = VISITING then return true
    if visited_tx[tid] = VISITED then return false
    visited_tx[tid] ← VISITING
    for each lock in transactions[tid].required_locks do
        for each holding_tid in lock.current_holders() do
            if holding_tid = tid then continue
            if DFS(holding_tid) then
                cycle.push(tid)
```

```
            return true
        end if
    end for
end for
visited_tx[tid] ← VISITED
return false
end procedure
```

When a lock conflict is encountered (i.e. a transaction tries to acquire a lock but is denied, or a permission check fails), the transaction is suspended and the relevant lock is added to the transaction's `required_locks` set. Control is then given back to the `Database` object, which calls `Database::on_control()` to initiate deadlock detection via Algorithm 2. If a victim is found, that transaction is rolled back to break the cycle, and the suspended transaction is resumed (provided it was not itself the victim). Note that this allows multiple transactions to be rolled back, since the resumed transaction can suspend again and trigger another round of deadlock detection.

b

i

The tuples themselves are stored in the `Relation` class's tuple container. We use a `StableVector` implemented as a linked list of fixed-size blocks. This allows us to keep stable pointers to tuples that can be stored in the various indices. Using a normal `std::vector` would require updating all pointers in the indices every time the vector resized, while using a plain linked list would lead to worse cache performance and allocation overhead.[3] Note that we only ever append tuples to the StableVector and never remove tuples from the middle; deletion is represented by setting the `alive` flag to false in the DataTuple.

Based on the query patterns discussed in 1(c), we maintain three kinds of hash indices on each relation. The first two are for queries of the form $R(c, Y)$ and $R(X, c)$, where we want to efficiently iterate over all tuples matching a constant on the left or right attribute, respectively. The third index is for queries of the form $R(X, X)$, where we want to efficiently iterate over all tuples whose left and right attributes are equal. Since each of these indices covers the same range as the predicate locks described above, we also place them in the `Group` class.

Note that we do not maintain a whole-relation index. We chose not to do this because it would increase tuple insertion overhead without providing a significant speed-up: we can already use the left/right indices to efficiently check whether a tuple exists. To iterate over all tuples in a relation, we directly scan the `StableVector` of tuples, which is more efficient than maintaining an additional index for this purpose.

ii

We first discuss the methods exposed by the `Relation` class:

`Relation::get_tuple(left: uint32_t, right: uint32_t) -> DataTuple*`
retrieves a pointer to the `DataTuple` with the given left and right attributes. It creates the tuple if necessary, so it never returns `null`. Note that it does not guarantee that the tuple is alive, so the caller should check the `alive` flag if they only want alive tuples. This works by using `l_to_r_index` to find

[3] In C++26, we could instead use a `std::hive`, but this is out of scope for this project.

the group of tuples with the given left attribute, and then using `Group::find` to find the tuple with the given right attribute within that group.

`Relation::edit_tuple(left: uint32_t, right: uint32_t) -> boolean`
is used to add or delete a tuple from the relation by setting its `alive` flag. It first finds the `DataTuple` via `Relation::get_tuple`, then checks the relevant locks to prevent phantoms and tries to acquire an exclusive lock on the tuple. If any lock check or acquisition fails, it records the locks the transaction needs and returns `false` to signal that the transaction should be suspended. Otherwise, it updates the tuple's alive status and returns `true`.

`Relation::begin() -> Relation::TupleContainer::iterator`
`Relation::end() -> Relation::TupleContainer::iterator`
return iterators that allow for iterating over all tuples in the relation. This is used for queries that need to scan the entire relation, such as $R(X, Y)$ with no constants.

Then we discuss the methods exposed by the `Group` class:

`Group::insert(tp: *DataTuple): unit` inserts the given tuple pointer into the group index. It is used when adding a new tuple to the relation and updating the indices.

`Group::find(left: uint32_t, right: uint32_t) -> DataTuple*` finds the tuple with the given left and right attributes in the group index. It returns `null` if no such tuple exists. It is used to efficiently locate tuples matching a given predicate when evaluating queries.

`Group::begin() -> Group::TupleContainer::iterator`
`Group::end() -> Group::TupleContainer::iterator`
return iterators that allow for iterating over all tuples in the group. This is used for queries that need to scan all tuples matching a certain predicate, such as $R(1, Y)$ or $R(X, 1)$.

c

Handling queries begins with parsing the input query into a vector of `QueryAtoms`, where each `QueryAtom` represents an atom in the query with its relation name and left/right arguments (which can be constants or variables). This is done in `Transaction::StartQuery`.

To handle suspensions cleanly, we use a volcano-style query evaluation algorithm, which naturally supports suspension between calls to `next()` since the objects in the query pipeline maintain their own local state.

The algorithm has two phases: setting up the query pipeline, and pulling tuples through the pipeline.

The query pipeline for a query $Q = R_1(l_1, r_1), ..., R_n(l_n, r_n)$ consists of $n + 1$ `Stages`. Stage 0 is always an initial stage while stage i for $i > 0$ corresponds to the effect of the query atom $R_i(l_i, r_i)$.

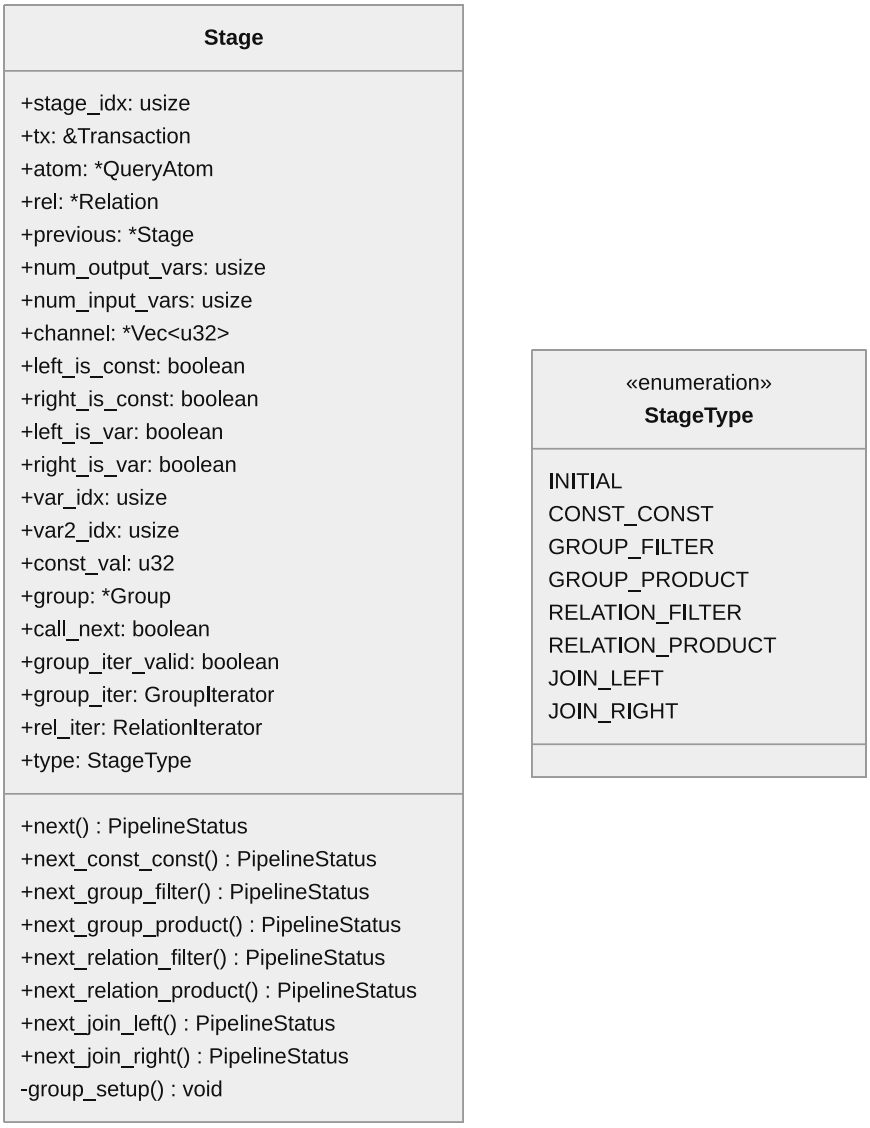

Figure 7: Stage Class

Each `Stage` (apart from the initial stage) is connected to a previous `Stage` via the `previous` pointer. When `next()` is called on a stage, it can pull tuples from the previous stage via `previous.next()` as many times as needed to produce the next output tuple. While stages conceptually pass tuples to each other, in practice no stage modifies variables already assigned by previous stages, so all stages share the same output buffer, `channel`.

Pipeline construction is initiated by `Transaction::StartQuery` when a query is first executed. This function constructs and connects all stages. Stage-specific configuration is handled by the `Stage` constructor, which determines the stage type from the query atom and previously seen variables, and acquires the relevant predicate locks.

Algorithm 3: Transaction::StartQuery

```
procedure Transaction::StartQuery(query: Vec<QueryAtom>)
            → StatusCode
  state ← EXECUTING_QUERY
  num_answers ← 0
```

```
    # Build variable-to-index mapping
    num_vars ← 0
    var_idx.clear()
    for each atom in query do
        if atom.relation ∉ relations then
            # Print warning message
            return StatusCode::FINISHED
        end if
        for each arg in {atom.left, atom.right} do
            if arg is Variable then
                if arg.name ∉ var_idx then
                    var_idx[arg.name] ← num_vars
                    num_vars ← num_vars + 1
                end if
            end if
        end for
    end for
    # Build query pipeline stages
    this.query_atoms ← query
    channel.resize(num_vars)
    stages.clear()
    for i ← 0 to query_atoms.len() do
        stages.push(Stage(i, *this))
    end for
    return resume_query()
end procedure
```

Algorithm 4: Stage::Stage

```
procedure Stage::Stage(stage_idx: usize, tx: &Transaction)
    if stage_idx = 0 then
        type ← INITIAL
        return
    end if
    atom ← tx.query_atoms[stage_idx - 1]
    previous ← tx.stages[stage_idx - 1]
    rel ← tx.relations[atom.relation]
    num_input_vars ← previous.num_output_vars
    match (atom.left, atom.right)
        (Constant(c1), Constant(c2)) ⇒
            type ← CONST_CONST
            num_output_vars ← num_input_vars
        (Constant(c), Variable(v)) ⇒
```

```
        var_idx ← tx.var_idx[v.name]; const_val ← c.value
        group ← rel.l_to_r_index[const_val]; group_setup()
    (Variable(v), Constant(c)) ⇒
        var_idx ← tx.var_idx[v.name]; const_val ← c.value
        group ← rel.r_to_l_index[const_val]; group_setup()
    (Variable(v1), Variable(v2)) ⇒
        idx1 ← tx.var_idx[v1.name]; idx2 ← tx.var_idx[v2.name]
        var_idx ← idx1; var2_idx ← idx2
        if idx1 = idx2 then
            group ← rel.diagonal_index; group_setup()
        else
            if idx1 ≥ num_input_vars and idx2 ≥ num_input_vars then
                type ← RELATION_PRODUCT; num_output_vars ← num_input_vars + 2
                tx.acquire(rel.whole_rel_lock, SHARED)
            else if idx1 ≥ num_input_vars then
                type ← JOIN_RIGHT; num_output_vars ← num_input_vars + 1
            else if idx2 ≥ num_input_vars then
                type ← JOIN_LEFT; num_output_vars ← num_input_vars + 1
            else
                type ← RELATION_FILTER; num_output_vars ← num_input_vars
                tx.acquire(rel.whole_rel_lock, SHARED)
            end if
        end if
    end match
end procedure
procedure Stage::group_setup()
    tx.acquire(group.lock, SHARED)
    if var_idx ≥ num_input_vars then
        type ← GROUP_PRODUCT
        num_output_vars ← num_input_vars + 1
    else
        type ← GROUP_FILTER
        num_output_vars ← num_input_vars
    end if
end procedure
```

The second phase of the query algorithm involves repeatedly pulling tuples through the pipeline by calling `Stage::next()` from the last stage until it either returns `SUSPEND` or `FINISHED`. This is orchestrated by `Transaction::resume_query()`.

Algorithm 5: Transaction::ResumeQuery

```
procedure Transaction::ResumeQuery() → StatusCode
    while true do
```

```
    match stages.last().next()
      OK ⇒
        num_answers ← num_answers + 1
        print_tuple(query_channel)
        continue
      SUSPEND ⇒
        return StatusCode::SUSPENDED
      FINISHED ⇒
        state ← READY
        print "Number of answers: ", num_answers
        return StatusCode::FINISHED
    end match
  end while
end procedure
```

`Stage::next()` matches on the stage type and calls one of the specific `next_*` functions. Each function implements the following transformation from input tuple set W to output tuple set W':

- `next_const_const()`: $W' = W$ if $R(c_1, c_2)$ holds, and $W' = \{\}$ otherwise.
- `next_group_filter()`:
  - For $R(x, c)$: $W' = \{t \in W \mid (t.x, c) \in R\}$
  - For $R(c, x)$: $W' = \{t \in W \mid (c, t.x) \in R\}$
  - For $R(x, x)$: $W' = \{t \in W \mid (t.x, t.x) \in R\}$
- `next_group_product()`:
  - For $R(x, c)$: $W' = \{(t, x) \mid t \in W, (x, c) \in R\}$
  - For $R(c, x)$: $W' = \{(t, x) \mid t \in W, (c, x) \in R\}$
  - For $R(x, x)$: $W' = \{(t, x) \mid t \in W, (x, x) \in R\}$
- `next_relation_filter()`: $W' = \{t \in W \mid (t.x_1, t.x_2) \in R\}$
- `next_relation_product()`: $W' = \{(t, x_1, x_2) \mid t \in W, (x_1, x_2) \in R\}$
- `next_join_left()`: $W' = \{(t, x_2) \mid t \in W, (t.x_1, x_2) \in R\}$
- `next_join_right()`: $W' = \{(t, x_1) \mid t \in W, (x_1, t.x_2) \in R\}$

These are implemented in volcano style in a mostly straightforward way:

- `next_const_const` checks whether the constant tuple is in R and forwards all or none of the tuples from the previous stage accordingly.
- `next_group_filter` and `next_relation_filter` call `previous.next()` and check whether the relevant part of the incoming tuple passes the filter criteria. If so, they forward the tuple; otherwise, they discard it and call `previous.next()` again until a tuple passes or the previous stage finishes.
- `next_group_product` and `next_relation_product` maintain an iterator over the group or relation, respectively, to construct the Cartesian product. For one input tuple, they can produce many output tuples until the iterator is exhausted; then they call `previous.next()` to get a new input tuple and reset the iterator.
- `next_join_left` and `next_join_right` are similar to `next_group_product`, but the group they iterate over depends on the current input tuple from the previous stage because they must match one attribute against that tuple.

All these methods need to remember to check for read permission from the relevant tuple-level locks. The join operations additionally need to acquire the relevant group locks when they move to a new group based on the input tuple from the previous stage.

If any call to `previous.next()` suspends or any lock check/acquisition fails, the method needs to suspend and return `SUSPEND` without advancing the state of the stage, so that when it is resumed, it can retry the same operation from the same point. The stages ensure the state is maintained by not advancing the iterators for products and joins and by using the `call_next` boolean for filters to indicate whether the next call should call `previous.next()` or not.

We show pseudocode for `Stage::next_relation_filter()`, `Stage::next_group_product()`, and `Stage::next_join_left()` as examples of how the different stage types are implemented. The other stage types are implemented similarly based on the logic described above.

Algorithm 6: Query Evaluation Methods

```
procedure Stage::NextJoinLeft() → PipelineStatus
  loop
    if ¬ group_iter_valid then
      match previous.next()
        OK ⇒
          group ← rel.l_to_r_index[channel[var_idx]]
          group_iter ← group.begin()
          tx.acquire(group.lock, SHARED) # never fails
          group_iter_valid ← true
        other ⇒ return other
      end match
    end if
    while group_iter ≠ group.end() do
      tp ← group_iter.second
      if ¬ tx.get_read_permit(tp.lock) then return SUSPEND
      group_iter ← group_iter.advance()
      if tp.alive then
        channel[var2_idx] ← tp.right
        return OK
      end if
    end while
    group_iter_valid ← false
  end loop
end procedure
procedure Stage::NextGroupProduct() → PipelineStatus
  loop
    if num_input_vars = 0 then
      # expected to be in group.begin() when set up
      # just 1 output tuple per group tuple
      match previous.next()
        OK ⇒
```

```
            if group_iter = group.end() then return FINISHED
        other ⇒ return other
    end match
  else if group_iter = group.end() then
    # expected to be in group.end() when set up
    match previous.next()
      OK ⇒ group_iter ← group.begin()
      other ⇒ return other
    end match
  end if
  tp ← group_iter.second
  if ¬ tx.get_read_permit(tp.lock) then return SUSPEND
  group_iter ← group_iter.advance()
  if tp.alive then
    if left_is_const then channel[var_idx] ← tp.right
    else channel[var_idx] ← tp.left
    return OK
  end if
end loop
end procedure
procedure Stage::NextRelationFilter() → PipelineStatus
loop
  if call_next then
    match previous.next()
      OK ⇒ pass
      other ⇒ return other
    end match
  end if
  tp ← rel.get_tuple(channel[var_idx], channel[var2_idx])
  if ¬ tx.get_read_permit(tp.lock) then
    call_next ← false
    return SUSPEND
  end if
  call_next ← true
  if tp.alive then return OK
end loop
end procedure
```

An alternative stage design is to use inheritance and a separate class for each stage type. This would reduce the number of state variables used for suspension. However, because different stage types share overlapping subsets of state, we found this decomposition unnecessarily complex. It would also require separate allocations instead of storing stages in a contiguous vector, leading to worse cache performance and more complex memory management.

A completely different approach would be an eager, set-at-a-time query evaluation algorithm. We chose not to use it because it would likely have worse memory performance (due to materialising large intermediate results) and is less natural to combine with suspension/resume semantics.

d & e

We use a single algorithm for both adding and deleting, with the `new_alive` parameter set to `true` for additions and `false` for deletions.

An important design consideration is that when a transaction fails either a lock permission check or a lock acquisition, as many relevant locks as possible are put into `required_locks`. This helps ensure deadlocks are detected as early as possible (without needing to resume transactions that cannot make meaningful progress).

Furthermore, we take care to avoid creating any new `DataTuple` objects within the relation if the transaction is not permitted by predicate locks, ensuring that containers to which suspended transactions may hold iterators are not modified.

We also store the original tuple values to allow rollbacks.

Algorithm 7: Single Tuple Edit Algorithm

```
procedure Relation::edit_tuple(tx: &Transaction, left: uint32_t, right: uint32_t,
            new_alive: bool) → boolean
  if ¬ permitted_by_pred_locks(tx.tid, left, right) then
    insert_required_locks_for_edit(tx, left, right)
    return false
  end if
  tp ← get_tuple(left, right) # creates the tuple if it doesn't exist
  if tx.acquire(tp->lock, EXCLUSIVE) then
    if tp->alive ≠ new_alive then
      tx.store_original(tp)
      tx.num_modified++
      tp->alive ← new_alive
    end if
    return true
  else
    insert_required_locks_for_edit(tx, left, right)
    return false
  end if
end procedure
procedure Relation::permitted_by_pred_locks(tx_id: TID, left: uint32_t, right:
uint32_t)
            → boolean
  if ¬ whole_rel_lock.permits_edit(tx_id) then return false
  if left = right and ¬ diagonal_index.lock.permits_edit(tx_id) then return false
  if ¬ l_to_r_index[left].lock.permits_edit(tx_id) then return false
  if ¬ r_to_l_index[right].lock.permits_edit(tx_id) then return false
  return true
```

```
end procedure
procedure Relation::insert_required_locks_for_edit(tx: &Transaction, left: uint32_t,
right: uint32_t) → unit
    # tp.lock already inserted via tx.acquire() if tp has been created
    tx.required_locks.insert(&whole_rel_lock)
    if left = right then
        tx.required_locks.insert(&diagonal_index.lock)
    end if
    tx.required_locks.insert(&l_to_r_index[left].lock)
    tx.required_locks.insert(&r_to_l_index[right].lock)
end procedure
```

f

On commit, we first ensure that the transaction is in the `READY` state and then release all held locks.

Algorithm 8: Commit Algorithm

```
procedure Transaction::commit() → StatusCode
    if state ≠ READY then
        print("ERROR: Trying to commit a transaction that is not ready.")
        return StatusCode::FINISHED
    end if
    for lock in held_locks do
        lock.release(tid)
    end for
    held_locks.clear()
    return StatusCode::FINISHED
end procedure
```

g

On rollback, we restore original tuple values and release all held locks. Note that unlike commit, we do not require the transaction to be in the `READY` state since we may need to roll back transactions that are currently suspended and waiting for locks.

Algorithm 9: Rollback Algorithm

```
procedure Transaction::rollback() → StatusCode
    for (tp, original_alive) in original_values do
        tp->alive ← original_alive
    end for
    for lock in held_locks do
        lock.release(tid)
    end for
    held_locks.clear()
    return StatusCode::FINISHED
```

end procedure

2

My implementation was developed on a laptop with a `11th Gen Intel(R) Core(TM) i7-11800H @ 2.30GHz`.

I developed it on NixOS with GCC `15.2.0` and confirmed that it compiles correctly on Ubuntu `24.04.4` with GCC `15.2.0-14ubuntu1~24~ppa1` via a Github Actions workflow using the `compile_nocmake.sh` script.

3

a

	Action	Time (ms)	# of Answers
Step 1	Import:	13610	
	Rollback:	3100	
Step 2	Import:	8269	
	Commit:	3060	
Step 3	Query 1:	0	4
	Query 2:	557	264
	Query 3:	0	6
	Query 4:	0	34
	Query 5:	0	719
	Query 6:	174	1048532
	Query 7:	0	67
	Query 8:	6	7790
	Query 9:	651	27247
	Query 10:	0	4
	Query 11:	0	224
	Query 12:	0	15
	Query 13:	0	472
	Query 14:	129	795970
	Commit:	15	
Step 4	Delete:	422	
	Rollback:	136	
Step 5	Delete:	489	
	Commit:	188	
Step 6	Query 1:	0	0
	Query 2:	0	0
	Query 3:	0	0
	Query 4:	0	0
	Query 5:	0	0

Action	Time (ms)	# of Answers
Query 6:	167	923871
Query 7:	0	0
Query 8:	2	4040
Query 9:	337	22522
Query 10:	0	0
Query 11:	0	186
Query 12:	0	9
Query 13:	0	413
Query 14:	119	701364
Commit:	14	

Table 2: Benchmarking Results

b

The results show that this implementation is better optimised for query-heavy workloads than for write-heavy workloads as query evaluation times are much lower than data import times. This is expected: edits require acquiring more locks, updating each transaction's `held_locks`, and maintaining multiple group indices. Importing many tuples at once also leads to multiple memory allocations which take time. By contrast, queries usually require fewer lock acquisitions because predicate locks can cover many tuples, and `SLocks` are always acquirable in shared mode.

There is a trade-off between these behaviours by increasing or decreasing index coverage. For example, we could add a whole-relation index, or create group locks without corresponding group indices. The current design is a balanced choice that avoids obviously poor performance for common query patterns.

In terms of scalability with respect to the number of open transactions, performance is good when contention is low. If transactions mostly touch different groups/tuples, they can acquire disjoint locks and run largely independently, so increasing the number of open transactions does not hurt throughput much.

The main scalability bottleneck appears once contention rises. More open transactions means larger lock-holder sets and more required-lock edges, and each suspension triggers a DFS deadlock check over that waits-for graph (cost proportional to $V + E$). In addition, coarse predicates such as $R(x, y)$ take relation-level shared locks, which can block many edits at once. So the system scales well in low-conflict workloads, but degrades under high-conflict workloads because more time is spent suspended or in deadlock handling rather than doing useful work; this is the expected trade-off for serializable isolation.

Finally, we note that hash-table layout is a major optimisation point in this project. We therefore use custom open-addressing hash tables with linear probing for the group indices, held locks, and log of original tuple values[4]. While these are better than the separate chaining hash tables provided by the C++ standard library, they are not as optimised as those from third-party libraries like Google's Abseil. The flame graph below shows that a large amount of time is spent in hash table operations, so using more optimised hash tables would likely lead to significant performance improvements.

[4]We don't replace the standard library hash table that maps left/right values to their groups since our implementations use buckets of keys and values. Large values like groups are bad for cache locality.

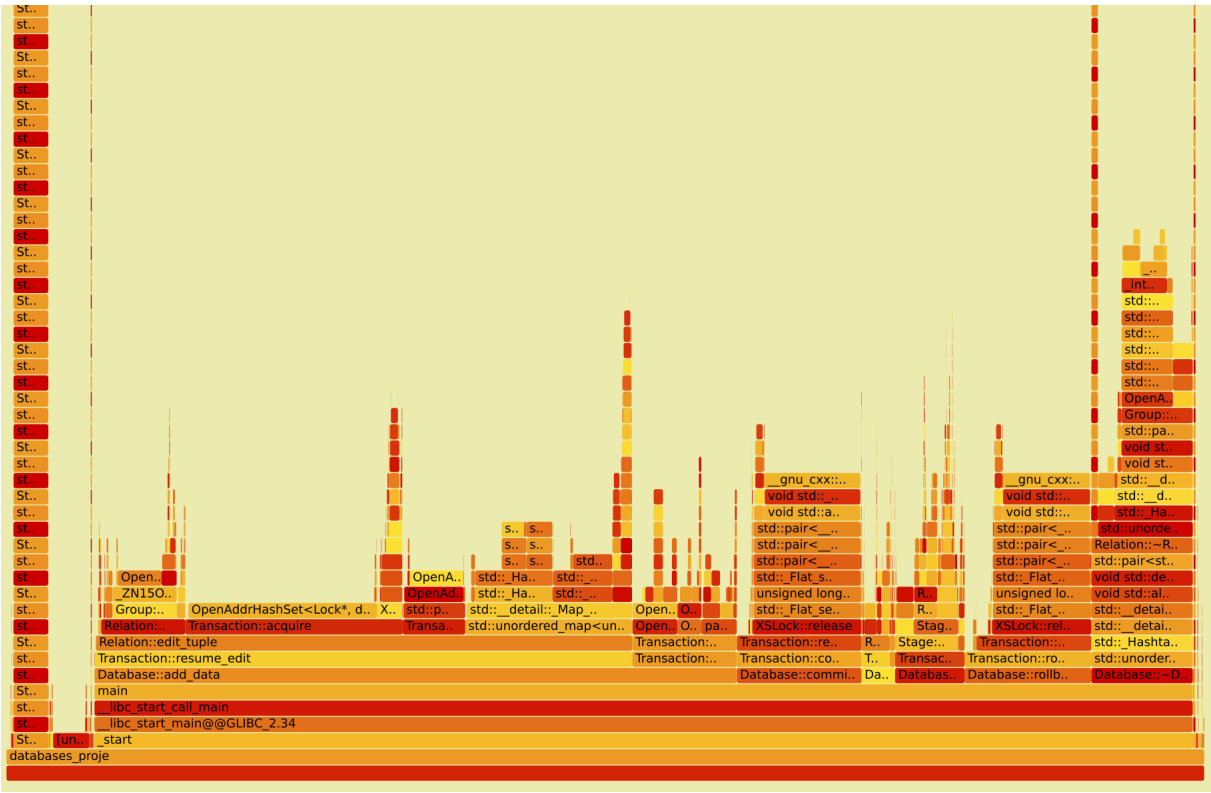

Figure 8: Flame Graph for the benchmarking procedure
Shows that a significant amount of time is still spent in hash table operations.